# Why should I trust Credit Scoring Models?

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***Abstract*—**

***Index Terms*—Naive Bayes (NB), Logistic Regression (LR), Bagged Decision Tree (BDT), Support Vector Machine (SVM), XGBoost (XGB), Ada Boost (ADB), Decision Tree Classifier (DTC), K-Nearest Neighbors (KNN), Random Forest (RF), Extra Tree Classifier (ETC), SHapley Additive exPlanations (SHAP), Local Interpretable Model-agnostic Explanations (LIME), Stepwise selection (STEP), Correlation (CORR), Neighborhood Rough Set (NRS), General Data Protection Regulation (GDPR), Equal Credit Opportunity Act (ECOA), Explainable Artificial Intelligence (XAI), Synthetic Minority Over-sampling Technique (SMOTE), Random Over-sampling (ROS), Random Under-sampling (RUS)**

## I. Introduction

Credit scores are numerical representations of a person's creditworthiness, derived from financial and personal data. They aid financial institutions in deciding whether to approve or reject loan applications. Lenders categorize clients into high-risk and low-risk groups, favouring low-risk borrowers with positive credit scores **[1]**. The credit score type is a primary indicator of a customer's financial health and repayment capacity. A higher credit score generally indicates a lower risk of default, while a lower score suggests a higher risk. Therefore, accurately classifying customers based on their credit scores can help financial institutions in a) Assessing creditworthiness, b) Reducing default risk, c) Streamlining the loan approval process, and d) Personalizing loan offers **[2]**. Manually assessing and classifying loan applicants based on their credit scores is time-consuming, resource-intensive, and prone to inaccuracies. To address these issues, banks are increasingly adopting machine learning algorithms for efficient application processing, improved customer satisfaction, and accurate assessment of customer creditworthiness **[3]**.

The debate over determining the optimal model for credit score classification continues with an ongoing effort to enhance predictive performance and address the challenges associated with identifying high-risk customers. The European Union General Data Protection Regulation (GDPR) stipulate that, for automated decisions like credit approvals, a "right to explanation" needs to be maintained [4]. Explainable Artificial Intelligence(XAI) algorithms address the issues of the black box nature of machine learning models that can help them plan arrangements and policies accordingly, thus preventing potential risks. The Basel Accord requires explanations for denied loan applications to ensure that transparency is maintained in automated decisions in the financial sector [5]. Hence this research aims to identify the best-suited model for classifying customer credit scores by conducting an extensive comparative analysis of top-performing classifiers. Additionally, it will address the black-box nature of these models by applying XAI techniques, such as LIME and SHAP, to uncover hidden patterns associated with high-risk customers and other attributes contributing to their potential defaulter status. Explaining credit score predictions is crucial for the finance sector to meet regulatory requirements, build customer trust, and make fair lending decisions.

### *Motivation*

Our motivation for this study stems from the challenges financial institutions face in selecting accurate and transparent credit scoring models. The need for fairness and precision in credit evaluations, along with the opacity of many models, drives us to find solutions that foster trust and promote equitable financial practices.

### *Contribution*

This study addresses the crucial issue of selecting the most effective credit scoring model for financial institutions. By comparing various models, we provide insights for more accurate and fair credit evaluations.

Using XAI techniques like LIME and SHAP, we enhance model transparency, helping customers understand factors influencing their credit scores. This transparency fosters trust and improves communication between financial institutions and their customers. Our findings guide the development of equitable credit policies and personalized financial plans, contributing to better financial health and inclusion.

## II. Related Research

### A. *Evolution of Credit Scoring Models*

The evolution of credit scoring models has focused on enhancing both predictive performance and interpretability. Many studies have conducted extensive comparative analyses to identify the optimal models

for credit scoring while addressing issues like class imbalance. For instance, **[6]** conducted a comparative analysis of DTC, LR, and RF on the highly imbalanced European Cardholders dataset, using SMOTE to tackle the imbalance. Results indicated that RF was the top performer. Similarly, **[3]** compared SVM, NB, and RF on the Loan Prediction dataset, which included features influencing personal and financial attributes, finding that L-SVM and RF outperformed the other classifiers. Additionally, **[7]** evaluated a broader range of base and ensemble classifiers on the Australian Credit Scoring dataset, including LR, DTC, SVM, KNN, MLP, NB, RF, ADB, XGB, and ETC. Their study revealed that SVM and ensemble models like RF, ETC, and BDT achieved the highest accuracies. However, the study lacked hyperparameter tuning and did not address the issue of class imbalance. Further, in the context of deep learning models, **[8]** highlighted the scarcity of research considering class imbalance issues in Internet finance. They proposed a model combining Deep Forest (DF) with five resampling methods: random over-sampling (ROS), random under-sampling (RUS), synthetic minority over-sampling technique (SMOTE), Tomek links, and SMOTE+Tomek. Experiments on four Internet financial credit scoring datasets demonstrated the effectiveness of the RUS-DF model among the others. In another study, **[2]** used SMOTE as their sampling strategy and proposed an advanced layered weighted and majority voting framework. They ranked seven classifiers and further enhanced the top five with AdaBoost. Their model outperformed baseline approaches on the German Loan Approval and Australian Credit Scoring datasets. Additionally, **[9]** introduced a Layered Stacking Ensemble that incorporated KNN, Decision Tree, and SVM as input classifiers, with Gradient Boosting as the primary classifier. This cost-effective solution aimed to provide a suitable model for small-scale businesses with limited resources. A unique approach to addressing limited data samples in credit scoring was presented by **[10]**. Using the Qianhai Credit Information dataset, they applied a transfer learning strategy with the SPY-transfer and SPY-TrAdaBoost models to tackle the cold start problem. Their SPY-TrAdaBoost + Cold Start (XGB) model outperformed the baseline cold start models.

Based on the related research, it can be concluded that ensemble models such as Random Forest (RF), AdaBoost, Gradient Boosting, and Extra Trees Classifier (ETC) consistently outperform baseline approaches in the credit scoring context. Additionally, sampling strategies like Synthetic Minority Over-sampling Technique (SMOTE), Random Over-sampling (ROS), and Random Under-sampling (RUS) effectively address class imbalance, further enhancing the performance of credit scoring models. The stacking approach has also demonstrated comprehensive results, especially when combined with heterogeneous credit scoring models.

### *B. XAI in Finance*

Various sectors within finance are exploring the implementation of explainable artificial intelligence (XAI) techniques, with risk management and portfolio optimization being among the first adopters. In risk management, one key application is default prediction, which involves forecasting the likelihood of borrowers defaulting on their loans. This process provides crucial insights for both customers and institutions, including explanations for loan rejections and identifying potential biases **[11]**. AI techniques process large volumes of data to make these predictions, but the lack of transparency in AI-generated recommendations can hinder adoption. XAI addresses these concerns by offering clear explanations for predictions, including feature contributions and interactions, which helps ensure fairness and enhances the usability of AI in the financial sector **[12], [13]**.

A research **[14]**, used SHAP to enhance the interpretability of credit card fraud detection models, identifying the most influential features. SHAP waterfall plots were employed to examine individual company explanations, revealing that features with large positive or negative values strongly influenced predictions. By bridging research gaps in model transparency, optimizing performance, and reducing risks through extensive classifier comparisons, the demand for accurate and interpretable credit scoring models can be better addressed as the number of loan applicants continues to grow.

XAI is crucial for regulatory compliance, risk management, customer trust, and bias mitigation. It helps financial institutions explain automated decisions to meet regulations like GDPR and ECOA **[4]**. In risk management, XAI enhances the transparency of risk models, aiding in better monitoring and mitigation. By clarifying decision-making processes and ensuring models predict correctly for the right reasons, XAI builds customer trust, enhancing retention and satisfaction **[15]**.

## III. METHODOLOGY

In this paper, the Credit Score Classification dataset was studied. The proposed methodology involves several key steps, as illustrated in the accompanying methodology flowchart **(see Figure 1)**.

First, missing values were imputed with the mean and outliers by normalization. To address the high dimensionality of input attributes, feature transformations were performed, and feature selection was employed using combined scores of mutual information gain and F-values to include relevant features. Further, to tackle the problem of class imbalance, random oversampling, and random undersampling techniques were applied.

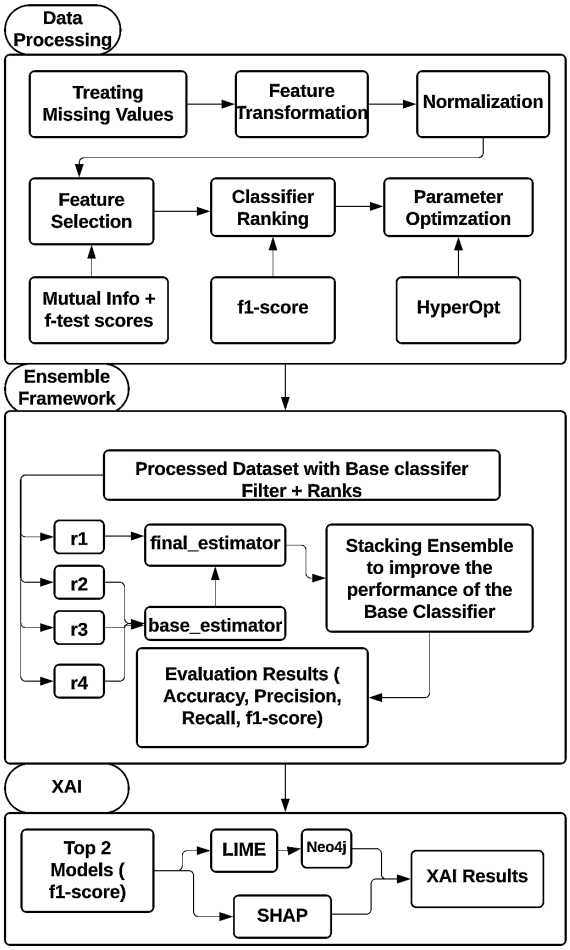


Fig. 1: Methodology

To identify the best classification model for the study, a comparative analysis of high-performing models across the credit score classification context, such as DTC, ETC, XGB, RF, KNN, and ADB, was conducted. A stacking ensemble was also proposed to improve overall classification performance. All models were evaluated based on precision, recall, f1-score, and accuracy.

Finally, to address the black-box nature of machine learning models in the credit scoring context, XAI techniques like LIME and SHAP were employed on the best-performing model to comprehensively explain captured patterns and anomalies affecting customer credit scores.

### A. *Dataset*

The *"Credit Score Classification"* **[16]** dataset on Kaggle aims to predict credit scores based on various features. It includes 100,000 records and 24 attributes related to applicants' demographics, financial history, and credit history, with the target variable *Credit Score* classifying scores into Good, Standard, and Poor. Despite being synthetic, the dataset aligns well with Experian's key credit scoring factors such as payment history, amounts owed, credit history length, and

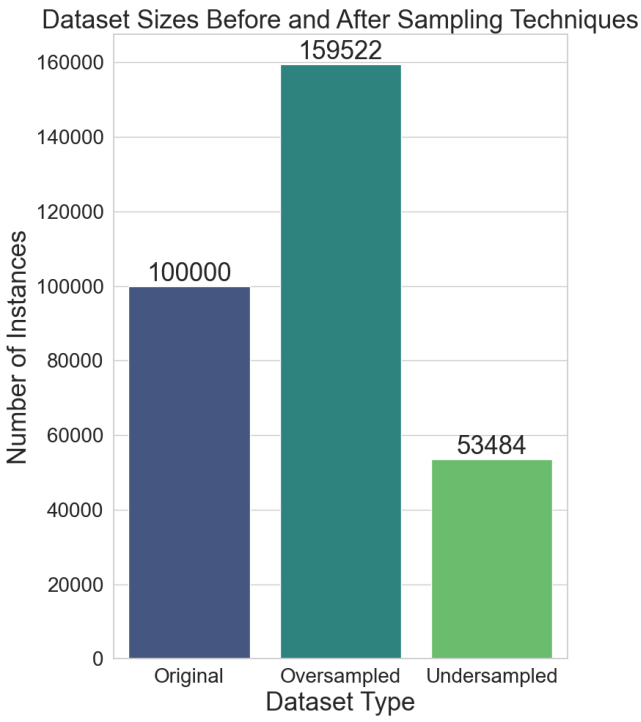


Fig. 2: Comparison of Dataset Sizes Before and After Sampling Techniques

credit mix **[17]**. These factors are represented by variables like *Delay_from_due_date*, *Outstanding_Debt*, *Credit_History_Age*, and *Num_Credit_Card*, providing a valuable foundation for model development and analysis.

### B. *Data preprocessing*

The data processing methodology undertaken in this study is detailed as follows:

To ensure the inclusion of only relevant features from the dataset attributes such as *ID*, *Customer_ID*, *SSN*, and *Name* were excluded due to their high cardinality. Subsequently, One-Hot Encoding was applied to nominal features, Ordinal Encoding to ordinal features, and numerical attributes were normalized.

To address the class imbalance present in the target variable *Credit Score*, classified as Good [17.83%], Standard [53.17%], and Poor [29%], both Random Oversampling and Random Undersampling techniques were employed. This study prefers Random Sampling over the commonly used SMOTE sampling in the context of Credit Fraud and Score classification, as noted in **[6], [18]**. The preference for Random Sampling stems from its use of real data points, which avoids the potential pitfalls of synthetic data, such as introducing noise and failing to accurately represent real-world scenarios **[18].**

In this study, Figure **2**, oversampling increased the *Credit Score* sample size to approximately 159,522, thereby enhancing the representation of the minority class. Conversely, undersampling reduced the sample

size to around 53,484, thereby balancing the class distribution.

Further, for feature selection, we utilized combined scores of k-mutual information gain and F-values, in addition to commonly used techniques in credit scoring such as STEP, CORR, t-test, and NRS **[2]**. This approach is adapted from **[9]**, where a combination of methods such as mean decrease Gini, information gain, and Chi-squared coefficient is employed to filter out relevant features for model training. The combined score $S$ for $k$-mutual information gains and $F$-values can be calculated using the following formula:

$$S = \sum_{i=1}^{k} (w_{1,i} \cdot \text{mutual_info_gain}_i + w_{2,i} \cdot \text{F_value}_i)$$

where:

- **S** is the combined score.
- **k** is the number of features considered.
- $w_{1,i}$ is the weight assigned to the $i$-th mutual information gain.
- $w_{2,i}$ is the weight assigned to the $i$-th $F$-value.
- $\text{mutual_info_gain}_i$ is the mutual information gain of the $i$-th feature.
- $\text{F_value}_i$ is the $F$-value of the $i$-th feature.

Combining mutual information gain and F-values captures both non-linear relationships and linear separability, providing a comprehensive evaluation of feature relevance. The top 15 features, based on these combined scores, were included in the model to ensure robust and efficient training.

### C. *Model Development*

In this study, the HyperOpt optimization framework **[19]** is used for model tuning in credit score classification, offering advantages over traditional GridSearch and Randomized Search approaches. Unlike these methods, which rely on exhaustive evaluations or random sampling of a fixed number of combinations, HyperOpt's Bayesian optimization adapts more effectively to variations in data distribution, particularly when dealing with oversampling and undersampling techniques. By continuously refining its search based on observed performance metrics, HyperOpt requires fewer evaluations to find optimal or near-optimal hyperparameters. This makes HyperOpt more efficient and scalable for large datasets or complex models.

Further to identify the most suitable model in the study, an extensive comparative analysis of top-performing models, as referenced in **[2], [3], [7], [9]**, was conducted. The models analyzed included DTC, ETC, XGB, RF, KNN, and ADB. These models were further optimized using the HyperOpt framework and ranked based on f1-scores. The top four ranked classifiers were then passed to a stacking ensemble, with the rank 1 model set as the final estimator and the rank 2, 3, and 4 models set as base estimators. Alongside the f1-score, other metrics such as precision, recall, and accuracy were used to evaluate model performance.

### D. *Explanability*

The top-performing model, assessed by its F1-score and accuracy, was further analyzed using LIME and SHAP to explain its predictions on the test dataset for each class. LIME-generated explanations were aggregated into a data frame and saved as a CSV for further analysis.

Use of Neo4j's graph database structure to visualize and analyze complex relationships between instances and features, as supported by **[20]**. A subset of 20 instances was sampled from LIME results. In the graph, each instance and feature was represented as a node, with relationships based on LIME values. All instance feature scores were treated as undirected and weightless relationships. NaN values indicated no relationship between nodes. To handle features with minimal impact, a threshold score of 0.01 was applied to all features, adjusting the sensitivity of relationship formation in the graph. Thresholds of 0.05 and 0.5 were also tested, but a threshold of 0.01 was found to capture the most interpretable graph in the context of feature importance in LIME predictions.

Using Neo4j Cypher queries, duplicate nodes were removed first. The APOC library's 'apoc.refactor.mergeNodes' procedure was utilized to merge these duplicates **[21]**, combining their properties and relationships. The graph was then projected using the Graph Data Science (GDS) **[22]** library, including Instance and Feature nodes with HAS_VALUE relationships.

Betweeness centrality is was utilized to highlight influential nodes among multiple entities in the Panama Papers data **[23]**, to measure the influence of a node by the number of shortest paths passing through it, highlighting features that play a critical role in the model's decisions. Betweenness Centrality algorithm was applied to the projected graph to calculate centrality scores for each node, determining its importance or influence within a network. The betweenness centrality $C_B(v)$ for a node $v$ is defined as:

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

where:

- $\sigma_{st}$ is the total number of shortest paths from node $s$ to node $t$.
- $\sigma_{st}(v)$ is the number of those shortest paths that pass through node $v$.

In the context of LIME feature importance, a feature with high betweenness centrality frequently connects different instances, indicating its crucial impact on model predictions. This measure aligns with LIME's objective of identifying pivotal features in model explanations, providing a comprehensive overview of feature importance across instances.

To elucidate the model's decision-making process, SHAP was employed to quantify feature importance.

SHAP values quantify each feature's marginal contribution to the prediction by considering all possible combinations of features (coalitions). SHAP computes the difference in the model's output when a feature is included versus excluded from a coalition. This marginal contribution is averaged across all possible coalitions, resulting in the SHAP value for that feature. By computing SHAP values for each instance in the test set, the relative contributions of features to the model's output were determined. A KernelExplainer was utilized to estimate these values, leveraging a background train dataset for comparison. This approach enabled a granular understanding of how features interacted to influence credit scores, transparently providing insights into the model's underlying logic for each class.

## IV. Results & Analysis

### A. *Model Results*

The performance of several machine learning models was evaluated using both oversampling and undersampling techniques to handle class imbalance. The models assessed include XGB, ETC, RF, KNN, DTC, and ADB. For each model, we used the f1-score and accuracy as the primary metric for ranking while ensuring a balanced evaluation of precision and recall. To improve prediction performance, the stacking ensemble approach was also employed alongside sampling techniques.

In the oversampling scenario before hyperparameter optimization, ETC and RF demonstrated the highest performance with weighted f1-scores of approximately 0.91. KNN and DTC performed moderately well, achieving weighted f1-scores of 0.86 and 0.90, respectively, while ADB lagged with a weighted f1-score of 0.71. Post optimization, the models showed significant improvements, with XGB achieving a weighted f1-score of 0.90, RF maintaining its top performance, and KNN improving to a weighted f1-score of 0.87. The stacking ensemble method, combining RF, XGB, KNN, and ETC, yielded an impressive weighted f1-score of 0.91, highlighting its efficacy in leveraging the strengths of individual models.

| Rank | Model | Precision | Recall | F1-Score | Accuracy |
|---|---|---|---|---|---|
| 1 | RF (Over) | 0.91 | 0.91 | 0.91 | 0.909857 |
| 2 | ETC (Over) | 0.91 | 0.91 | 0.91 | 0.909420 |
| 3 | DTC (Over) | 0.90 | 0.90 | 0.90 | 0.897916 |
| 4 | **Stacking (Over)** | 0.90 | 0.90 | 0.89 | 0.883111 |
| 5 | KNN (Over) | 0.86 | 0.86 | 0.86 | 0.858674 |

TABLE I: Top 5 Classifiers Before HyperOpt

In the undersampling scenario, before optimization, RF and ETC emerged as top performers, achieving weighted f1-scores of 0.82 and 0.81, respectively. KNN and DTC had moderate performance with weighted f1-scores of 0.77 and 0.76, respectively, while ADB remained the least effective with a weighted f1-score of 0.71. After optimization, XGB showed notable improvement with a weighted f1-score of 0.82, matching the performance of RF. ETC and KNN also demonstrated slight improvements in their weighted f1-scores, achieving 0.81 and 0.78, respectively. The undersampling stacking ensemble, combining RF, XGB, KNN, and DTC, maintained a solid performance with a weighted f1-score of 0.82. Top 5 classifiers before are highlighted in **I** and post-optimization are highlighted in **II**

| Rank | Model | Precision | Recall | F1-Score | Accuracy |
|---|---|---|---|---|---|
| 2 | RF (Over) | 0.91 | 0.91 | 0.91 | 0.909857 |
| 3 | ETC (Over) | 0.91 | 0.91 | 0.91 | 0.909544 |
| 1 | **Stacking (Over)** | 0.91 | 0.91 | 0.91 | 0.910000 |
| 4 | XGB (Over) | 0.90 | 0.90 | 0.89 | 0.895784 |
| 5 | KNN (Over) | 0.87 | 0.87 | 0.86 | 0.867513 |

TABLE II: Top 5 Classifiers After HyperOpt

The stacking ensemble method, particularly in the oversampling context, provided the best overall performance. This method not only improved the accuracy but also ensured balanced precision and recall across all classes, making it a suitable choice for further analysis and explainability using XAI techniques.

### B. *Explanability Results*

The findings from XAI align with the established framework of credit scoring, as outlined by FICO **[24]**. Key determinants of creditworthiness, such as payment history, credit utilization, and credit history length, were shown to be significant predictors in the model, corroborating their importance as per industry standards.

For the Poor class, as shown in ***Figure 3***, the SHAP analysis identifies several critical factors: high interest rates and high delays from due dates strongly impact poor ratings positively, while low outstanding debt and low delays from due dates contribute negatively to poor credit scores. Shorter credit histories and lower incomes increase the impact of poor classifications, while the credit mix has a limited effect. Additionally, the number of credit inquiries and monthly in-hand salary exhibit a neutral impact.

The LIME centrality analysis for the Poor class, detailed in (**III**), supports these findings by highlighting the most influential nodes in the network for the selected sample: 'Outstanding Debt' is the most influential, followed by 'Number of Credit Inquiries' and 'Interest Rate.' These features with high centrality are located in the same vicinity within the graph structure in ***Figure 3***, collectively impacting the chosen sample of LIME instances to be classified as 'Poor'.

***Figure 5*** shows analysis of 'Standard' credit scores which indicated that a high credit mix negatively impacts predictions, contrasting sharply with its positive influence on good credit scores. This suggests that the credit mix serves as a critical transition point between

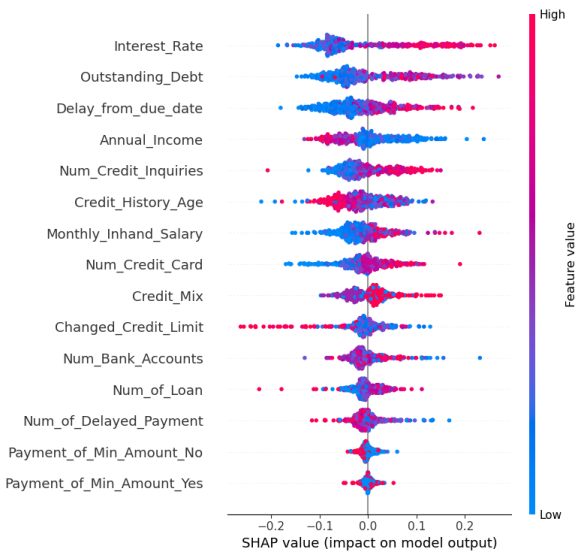


Fig. 3: SHAP Summary Plot: Key Drivers of 'Poor' Credit Score Predictions

Fig. 4: Neo4j Graph for Credit Score = Poor Class for 20 Instances

standard and good credit scores. High interest rates consistently detract from both good and standard credit scores, indicating that higher borrowing costs correlate with increased risk. In contrast, low and moderate outstanding debt levels show a neutral to slightly positive impact, with low debt being somewhat favourable, although its overall influence remains minimal. Delays from the due date have a moderate and variable effect, while higher annual incomes positively influence standard scores, albeit modestly. High credit inquiries negatively affect the likelihood of being classified as standard, whereas low inquiries slightly enhance it.

| Name | Centrality |
|---|---|
| Outstanding Debt | 91.798389 |
| Number of Credit Inquiries | 84.980349 |
| Interest Rate | 71.183954 |
| Number of Credit Cards | 37.076142 |
| Delay from Due Date | 36.398128 |
| Monthly Inhand Salary | 16.027198 |
| Annual Income | 4.292487 |
| Number of Loans | 0.706316 |
| Credit History Age | 0.537037 |
| Credit Mix | 0.000000 |
| Number of Bank Accounts | 0.000000 |
| Payment of Minimum Amount (No) | 0.000000 |
| Payment of Minimum Amount (Yes) | 0.000000 |
| Number of Delayed Payments | 0.000000 |
| Changed Credit Limit | 0.000000 |

TABLE III: Betweenness Centrality Scores for the Poor Class

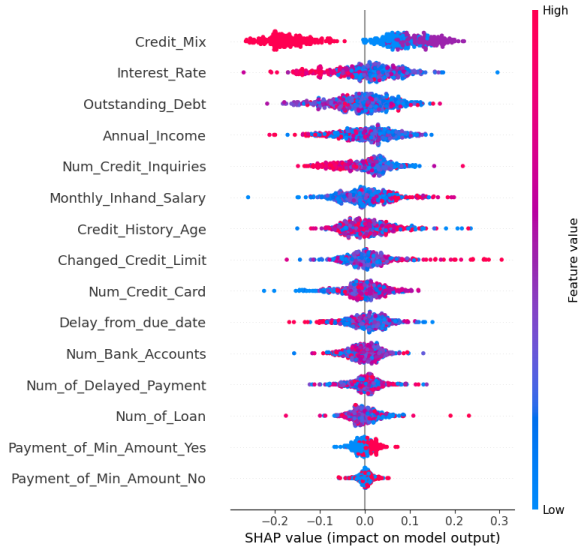


Fig. 5: SHAP Summary Plot: Key Drivers of 'Standard' Credit Score Predictions

The centrality analysis for the Standard class, **Table IV**, supports these findings, as features like Interest Rate, Delay from Due Date, Outstanding Debt, and Credit Mix share the highest centrality score. The validation of SHAP observations is evident in the LIME results, where high interest rates and delays from due dates align with their negative impacts on standard scores for the chosen sample. Additionally, the minimal influence of outstanding debt in both analyses suggests that while it may be a factor, it does not significantly impact standard credit scores, suggesting its polarizing role in 'Good and 'Poor' predictions.

For the Good class, ***Figure 6***, high credit mix positively impacts scores, while high interest rates have the opposite effect. High outstanding debt negatively impacts scores. Low delays from due dates positively impact scores, whereas longer delays have a slight negative effect. Higher incomes consistently relate to

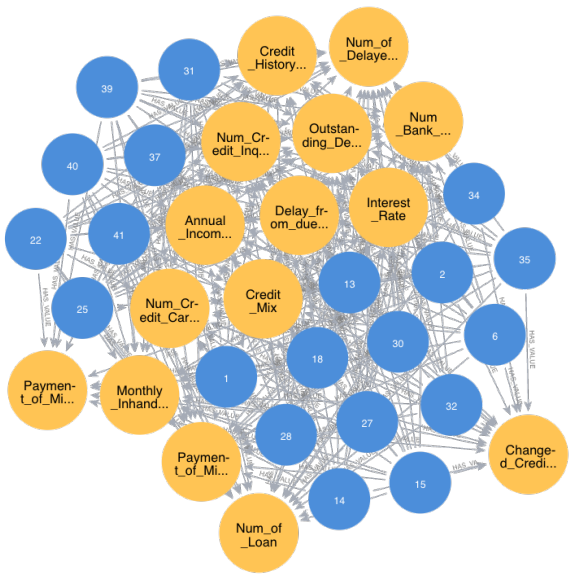


Fig. 6: Neo4j Graph for Credit Score = Standard Class for 20 Instances

Fig. 7: SHAP Summary Plot: Key Drivers of 'Good' Credit Score Predictions

| Name | Centrality |
|---|---|
| Interest Rate | 22.745868 |
| Delay from Due Date | 22.745868 |
| Outstanding Debt | 22.745868 |
| Credit Mix | 22.745868 |
| Monthly Inhand Salary | 22.745868 |
| Number of Credit Cards | 22.745868 |
| Number of Credit Inquiries | 22.745868 |
| Annual Income | 19.269308 |
| Number of Loans | 16.843298 |
| Credit History Age | 16.285197 |
| Number of Bank Accounts | 12.020336 |
| Payment of Minimum Amount (No) | 10.611333 |
| Number of Delayed Payments | 8.170952 |
| Changed Credit Limit | 2.467795 |
| Payment of Minimum Amount (Yes) | 0.110708 |

TABLE IV: Betweenness Centrality Scores for the Standard Class

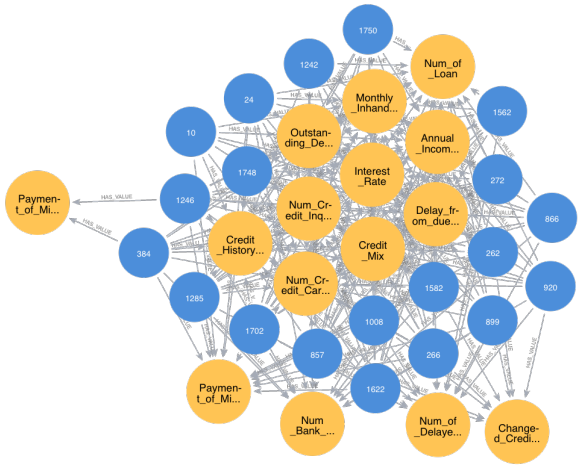


Fig. 8: Neo4j Graph for Credit Score = Good Class for 20 Instances

better scores, while lower incomes have a moderate negative impact. Interestingly, higher monthly in-hand salaries negatively affect score predictions.

The Good class centrality analysis, **Table V**, reveals that features like Interest Rate, Delay from Due Date, Outstanding Debt, Credit Mix, Monthly Inhand Salary, Number of Credit Cards, Payment of Minimum Amount (No), and Number of Credit Inquiries have the highest centrality scores. These features align with the SHAP plot, highlighting their consistent importance in both global and local explanations.

To explore how these central features interact with counter-intuitive findings, we examine LIME explanations for instances with high monthly in-hand salaries. **Fig. 9** LIME explanations show that while high monthly salary negatively impacts the score, factors such as high annual income, good credit mix, and a low number of credit cards significantly enhance the positive impact for instances classified as 'Good'. The feature scores from LIME for these factors contrast with the monthly in-hand salary, and their combined higher scores towards a positive prediction explain the model's inconsistency. This suggests that the identified central features collectively contribute positively to the prediction of good scores, outweighing the negative impact of high monthly salaries.

Monthly Salary and Annual Income both indicate a person's financial capability, but they have opposite effects on class prediction. These effects are particularly prominent in the 'Good' class, where the model places more weight on Annual Income than Monthly Salary. Possible explanations include inherent data bias towards Annual Income, the fluctuation of Monthly Salary in certain occupations, and the influence of factors such as the number of dependents or other liabilities on financial behavior.

| Name | Centrality |
|---|---|
| Interest Rate | 17.158917 |
| Delay from Due Date | 17.158917 |
| Outstanding Debt | 17.158917 |
| Credit Mix | 17.158917 |
| Monthly Inhand Salary | 17.158917 |
| Number of Credit Cards | 17.158917 |
| Payment of Minimum Amount (No) | 17.158917 |
| Number of Credit Inquiries | 17.158917 |
| Annual Income | 15.357406 |
| Number of Bank Accounts | 15.157195 |
| Number of Loans | 13.405994 |
| Number of Delayed Payments | 11.775008 |
| Credit History Age | 10.485905 |
| Changed Credit Limit | 9.052426 |
| Payment of Minimum Amount (Yes) | 3.494727 |

TABLE V: Betweenness Centrality Scores for the Good Class

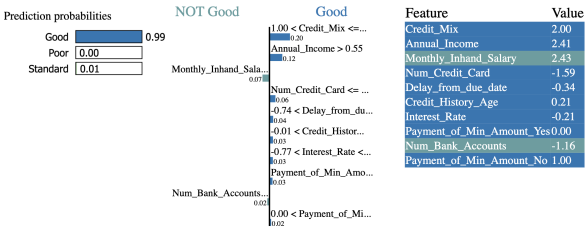


Fig. 9: This depicts a random instance of 'Good' predicted class from the top 10 per cent quartile of the feature Monthly_inhand_salary.

## V. Conclusions

In this study, we have demonstrated the comparative analysis of top performing credit scoring models across the context of credit scoring alongside the proposed stacking ensemble model which revealed to be the top-performing model for the dataset taken into the study, particularly when augmented with oversampling techniques. The stacking ensemble's ability to integrate multiple base models resulted in improved accuracy with balanced precision and recall making it a robust tool for predicting creditworthiness.

Additionally, the use of XAI techniques such as LIME and SHAP has provided valuable insights into the decision-making process of the ensemble model. These techniques allowed for a detailed understanding of how features like 'Outstanding Debt', 'Interest Rate', 'Credit Mix' and 'Annual Income' contributed to credit predictions as well as analysed counter-intuitive explanations for Monthly Salary by complementing SHAP with LIME paired with Graph Analytics, ensuring that the stacking model predicted credit scores for the right reasons and aligned with well-established credit-scoring frameworks as mentioned in **[17], [24]**.

Overall, the findings of this study have significant implications for financial institutions. By adopting advanced machine learning models and XAI techniques in conjunction with Graph Data Science, institutions can develop fairer and more transparent credit policies for both regulatory compliance and maintaining customer trust, as it allows for clear and justifiable credit approval or denial explanations. The ability to provide clear explanations for credit decisions not only improves customer relationships but also ensures adherence to regulatory standards, fostering a more equitable and trustworthy credit market.

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